Lesson 7: How Powerful Are You?

Unit

Energy: Stores and Transfers

Lesson focus

Students learn that power is the rate of energy transfer, calculate it with P = E ÷ t, rearrange to find energy or time, and measure their own power by walking up a flight of stairs.

Specification

AQA GCSE Physics 4.1.1.4 Power. The equation is on the AQA Physics equation sheet. The specification also gives power = work done ÷ time; this unit uses P = E ÷ t only, and work done (W = F s) is left to the Forces unit.

Scope

Students calculate P, and rearrange to find E (E = P t) or t (t = E ÷ P).

Learning objectives

By the end of the lesson, students should be able to:

1. Define power as the rate at which energy is transferred.
2. State that 1 watt is 1 joule per second.
3. Calculate power using P = E ÷ t, and rearrange it to find E or t.
4. Convert between W and kW, J and kJ, and minutes and seconds.
5. Explain why two motors lifting the same load through the same height can have different powers.

Key vocabulary

Keyword	Student-friendly definition
Power	How fast energy is transferred; the energy transferred each second
Watt (W)	The unit of power; 1 W = 1 J/s
Kilowatt (kW)	1000 W
Rate	How much happens per second

The equation

Quantity	Symbol	Unit
Power	P	watt (W)
Energy transferred	E	joule (J)
Time	t	second (s)

$$P = E \div t$$

power = energy transferred ÷ time

Rearranged: E = P × t, and t = E ÷ P.

Equipment and resources

- Two toy cranes or a small electric motor lifting a load at two settings (or a video of two lifts)
- Stopwatches, a metre rule or tape measure
- A flight of stairs with a known vertical height (measure one step and count the steps beforehand)
- Calculators

Safety notes (stair practical)

- Walk briskly; do not run. Hold the handrail.
- One student on the stairs at a time; others time from the top or bottom.
- Choose a dry, clear staircase and agree a route that avoids other classes.
- Students who cannot use stairs time and record.

Starter: Same job, different power

Two motors each lift a 2.0 kg load through 1.5 m. Motor A takes 2 s; motor B takes 6 s.

Ask: which transferred more energy? (Neither: both transferred 2.0 × 9.8 × 1.5 = 29.4 J to the gravitational potential store.) So what is different? (Motor A did it faster.)

> *Power tells you how fast energy is transferred, not how much.*

Main teaching: Calculating power

Worked example 1 (the starter):

> *P = E ÷ t*
> *Motor A: P = 29.4 ÷ 2 = 14.7 W*
> *Motor B: P = 29.4 ÷ 6 = 4.9 W*

Worked example 2 (finding energy):

> *A 2.0 kW kettle is switched on for 2 minutes. Calculate the energy transferred.*
> *P = 2.0 kW = 2000 W; t = 2 min = 120 s*
> *E = P t*
> *E = 2000 × 120*
> *E = 240 000 J*

Worked example 3 (finding time):

> *A 1500 W motor must transfer 45 000 J. How long does it take?*
> *t = E ÷ P*
> *t = 45 000 ÷ 1500*
> *t = 30 s*

Remind students that the power rating on an appliance (a 60 W lamp, a 2 kW kettle) tells you the energy it transfers every second.

Practical: How powerful are you?

1. Measure the vertical height of the flight of stairs: height of one step × number of steps.
2. Use a mass of 50 kg for everyone (see teacher notes).

3. Each student walks briskly up the stairs while a partner times them. Repeat if time allows.
4. Calculate the energy transferred to the gravitational potential store: E = m g h.
5. Calculate power: P = E ÷ t.

Sample results: 18 steps of 0.17 m, so h = 3.06 m; E = 50 × 9.8 × 3.06 = 1499 J ≈ 1500 J; time 5.0 s; P = 1500 ÷ 5.0 = 300 W.

Discussion:

1. Is your real power output higher or lower than your answer? (Higher: you also transferred energy to your kinetic store and to the thermal store of your body and the air.)
2. Compare with appliances: a lamp (10 W), a kettle (2000 W), a car (about 100 000 W).

Activity: Calculation page

Workbook page: straight P = E ÷ t; finding E and t; unit conversions (kJ, kW, minutes); comparing two machines that do the same job.

Check for understanding

1. What is power?
2. What is 1 W?
3. A lamp transfers 600 J in 10 s. What is its power? (60 W)
4. A 40 W fan runs for 5 minutes. How much energy does it transfer? (12 000 J)
5. Two students of the same mass climb the same stairs. One takes 4 s, the other 8 s. Compare their powers.

Plenary

"A more powerful car has more energy." True or false? Students explain. (False as stated: a more powerful car transfers energy faster. It may or may not have more energy stored in its fuel.)

Possible prep

Find the power rating on three appliances at home. Calculate the energy each transfers if it runs for 10 minutes.

Teacher notes

1. **Do not weigh students.** Using a standard 50 kg avoids embarrassment and safeguarding concerns; students who want to use their own mass can do so privately at home. If you use real masses, never share them aloud.
2. Minutes to seconds is the commonest error. So is kW to W.
3. Students often think power is the same as energy, or the same as force. Keep returning to "per second".
4. The specification's second equation, P = work done ÷ time, means the same thing once students meet work done in Forces. Mention it only if asked.